

$$C = A \times A$$

Q1) Approach

for $A^2 \rightarrow 1000 \times 1000 \times 0.1 \text{ A} \leftarrow 0.1$

$$A = \begin{pmatrix} A_{00} & A_{01} \\ A_{10} & A_{11} \end{pmatrix}$$

1 2	3 4
5 6	7 8
9 10	11 12
13 14	15 16

Tiles Multiplication

Split large Matrix

into small. Multiply & Add them

$$\begin{bmatrix} 158 & 174 \\ 374 & 422 \end{bmatrix} = 10$$

$$C = A \times A$$

$$C = \begin{pmatrix} C_{00} & C_{01} \\ C_{10} & C_{11} \end{pmatrix}$$

$A \times A \leftarrow$

$$C_{00} = A_{00} \times A_{00} + A_{01} \times A_{10}$$

$$\begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix} \times \begin{bmatrix} 9 & 10 \\ 13 & 14 \end{bmatrix}$$

$$\begin{bmatrix} 11 & 14 \\ 35 & 46 \end{bmatrix} + \begin{bmatrix} 79 & 86 \\ 167 & 182 \end{bmatrix} = \begin{bmatrix} 90 & 100 \\ 202 & 228 \end{bmatrix}$$

$$C_{01} = A_{00} \times A_{01} + A_{01} \times A_{11}$$

$$\begin{bmatrix} 158 & 174 \\ 374 & 422 \end{bmatrix}$$

$$C_{10} \rightarrow A_{10} \times A_{00} + A_{11} \times A_{10}$$

$$306 \quad 356$$

$$690 \quad 804$$

$$C_{11} = A_{10} \times A_{01} + A_{11} \times A_{11}$$

$$C_{11} = \begin{bmatrix} 374 & 422 \\ 854 & 966 \end{bmatrix}$$

$$C \rightarrow A \times A$$

$$100$$

$$000$$

$$= 0$$

$$110$$

$$010$$

$$\begin{bmatrix} C_{00} & C_{01} \\ C_{10} & C_{11} \end{bmatrix} = \begin{bmatrix} 90 & 100 & 158 & 174 \\ 202 & 228 & 374 & 422 \\ 306 & 356 & 598 & 646 \\ 418 & 484 & 822 & 886 \end{bmatrix}$$

$$\begin{bmatrix} 001 & 011 \\ 100 & 110 \end{bmatrix} = \begin{bmatrix} 28 & 11 \\ 128 & 28 \end{bmatrix} + \begin{bmatrix} 11 & 11 \\ 28 & 28 \end{bmatrix}$$

$$11A \times 10A + 10A \times 00A = 110$$

$$\begin{bmatrix} 111 & 821 \\ 281 & 118 \end{bmatrix}$$

Q2 Gaussian Elimination

necessary conditions.

$$Ax = B$$

A should be invertible.

$$x = BA^{-1}$$

Computing A^{-1} is very costly so we use gaussian elimination

Time complexity $\rightarrow O(n^3)$

Matrix

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{bmatrix}$$

$\rightarrow$ Symmetric $\rightarrow n \times n$
 $\rightarrow$ It should follow SDD for gaussian elimination

$$\rightarrow (i) |a_{ii}| > 0$$

$$|\Delta| > 0$$

Element in Matrix $\rightarrow [-10, 10]$

$$B \rightarrow \text{All } 100 \text{ } [1 \text{ to } 100]$$

$n \rightarrow$ range up 1000×1000 or 2000×2000

$$Ax = B.$$

$$A = \begin{pmatrix} a_{11} & - & - & a_{1n} \\ a_{21} & & & \vdots \\ \vdots & & & \vdots \\ a_{m1} & & & a_{mn} \end{pmatrix}$$

$$[A|B]$$

Forward Elimination. [into upper triangular]

$$m_{ik} = \frac{a_{ik}}{a_{kk}}$$

$$MA = U \Rightarrow Ax = b \Rightarrow MAx = Mb \Rightarrow Ux = Mb = \tilde{b}$$

$$MA = U \text{ block } U \leftarrow$$

elimination

$$i = k+1 - \dots - n$$

$$l_{ik}^{(k)} = \frac{a_{ik}^{(k)}}{a_{kk}^{(k)}}$$

$$\underline{LHS} \quad R_i \leftarrow R_i - l_{ik}^{(k)} R_k$$

$$a_{ik}^{(k+1)} = 0$$

RHS

$$b_i^{k+1} = b_i^k - l_{ik}^{(k)} b_k^{(k)}$$

Repeat for Every coloum.

$$k = 1, 2, \dots, n-1$$

$$A^n = E^{n-1} P^{n-1} \dots E^{(1)} P^{(1)} A = U$$

$$MA = U$$

$$Mb = \tilde{b}$$

$$Ux = \tilde{b}$$

$$\begin{matrix} 12 & 18 \\ 11 & 15 \end{matrix}$$

$$\forall i \rightarrow$$

$$\begin{matrix} 3 & 11 \end{matrix}$$

$$\begin{matrix} 1 & 5 \end{matrix}$$

$$\forall i \rightarrow$$

Back Substitution

$$x_n = \frac{\tilde{b}_n}{u_{nn}}$$

$$i = n-1$$

$$x_i = \frac{\tilde{b}_i - \sum_{j=i+1}^n u_{ij} x_j}{u_{ii}}$$

$$2011 + 1040 + 11u_{ii} = 0 \text{ a } \odot$$

$$i = n-1 \dots 1$$

$$\begin{matrix} 13 & 14 & 12 & 18 \\ 1 & \boxed{x} & 11 & 15 \\ 2 & 1 & 1 & 2 \\ 3 & 1 & 1 & 5 \end{matrix}$$

Complete approach

Stepwise

$$Ax = b$$

↓

Pivoting

↓

Calculate multipliers.

↓

$$\text{Elimination} \rightarrow MA = U \rightarrow Ux = b \rightarrow$$

Backward Substitution

$$B = \begin{matrix} \begin{matrix} 1 & 8 \\ 2 & 1 \end{matrix} & \begin{matrix} 1 & 8 \\ 2 & 1 \end{matrix} \\ \begin{matrix} 3 & 11 \\ 1 & 5 \end{matrix} & \begin{matrix} 3 & 11 \\ 1 & 5 \end{matrix} \end{matrix}$$

$$\begin{matrix} 2011 + 1040 + 11u_{ii} = 0 \text{ a } \odot \\ 11u_{ii} = -2011 - 1040 \\ u_{ii} = \frac{-2011 - 1040}{11} \end{matrix}$$